Thistle centreboard: the eight sections in thistle-screen

True scale. Chord 18 in (457.2 mm) square to the leading edge, 1 in (25.4 mm) thick: t/c 5.56 %. Section as cut square to the leading edge.

NACA 0006
max at 30 %, nose r 1.5 mm, TE 7.7°

SD8020
max at 29 %, nose r 0.8 mm, TE 4.9°

HT35
max at 26 %, nose r 2.0 mm, TE 5.5°

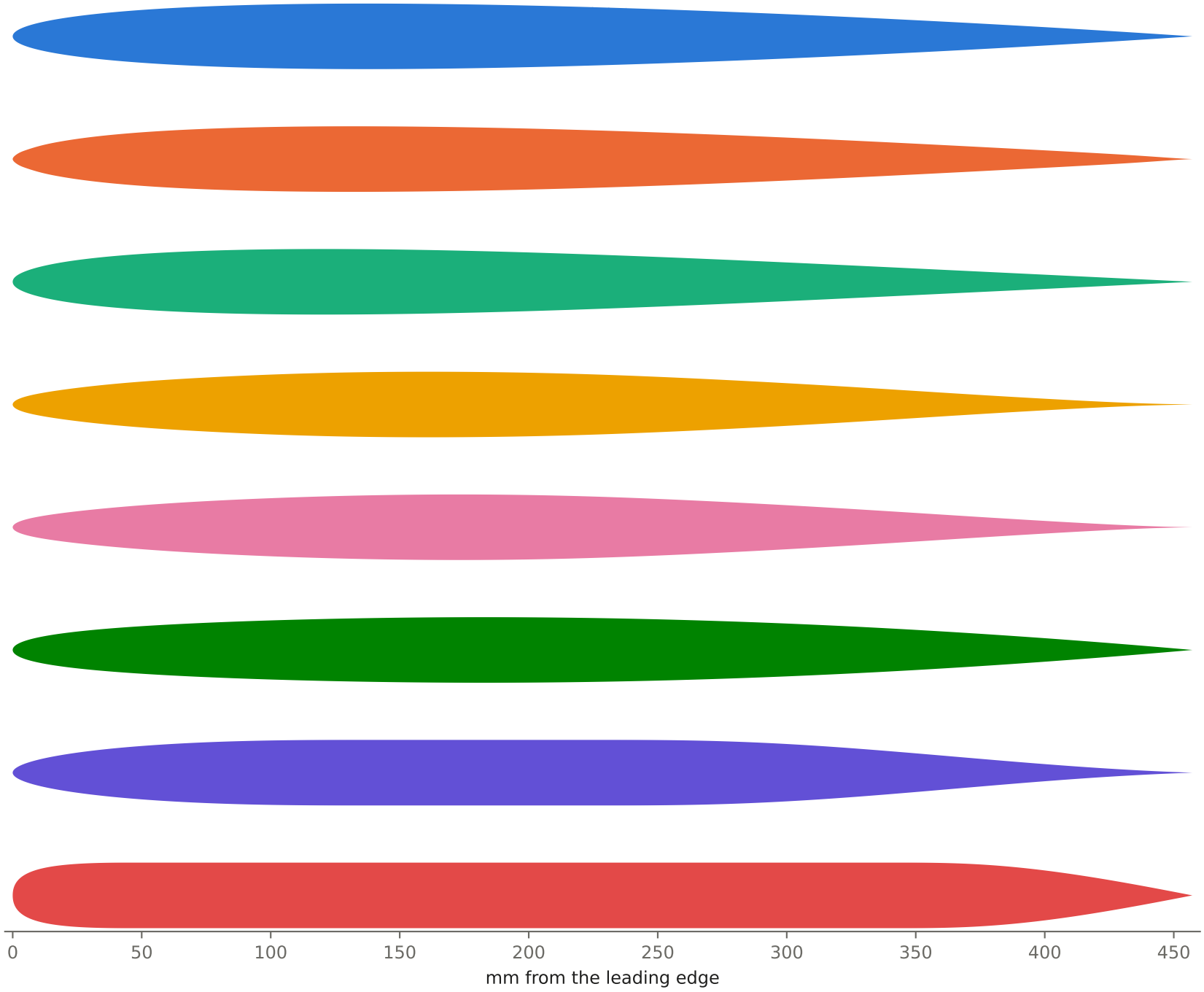
NACA 63-006
max at 35 %, nose r 1.0 mm, TE 2.2°

NACA 64-006
max at 38 %, nose r 1.0 mm, TE 1.8°

NACA 0006-64
max at 40 %, nose r 1.5 mm, TE 10.0°

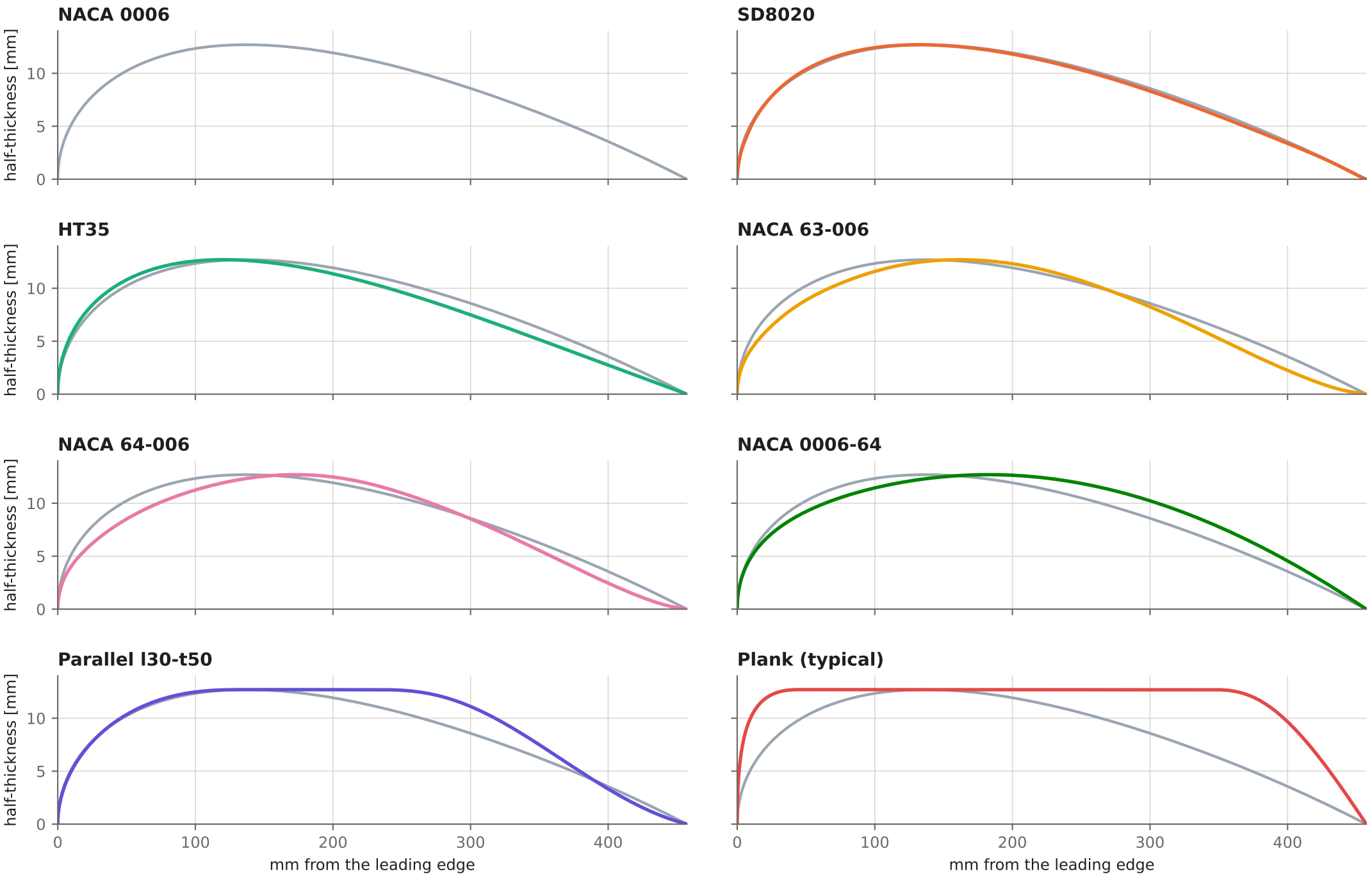
Parallel l30-t50
max at 50 %, nose r 1.3 mm, TE 3.9°

Plank (typical)
max at 75 %, nose r 12.7 mm, TE 22.1°



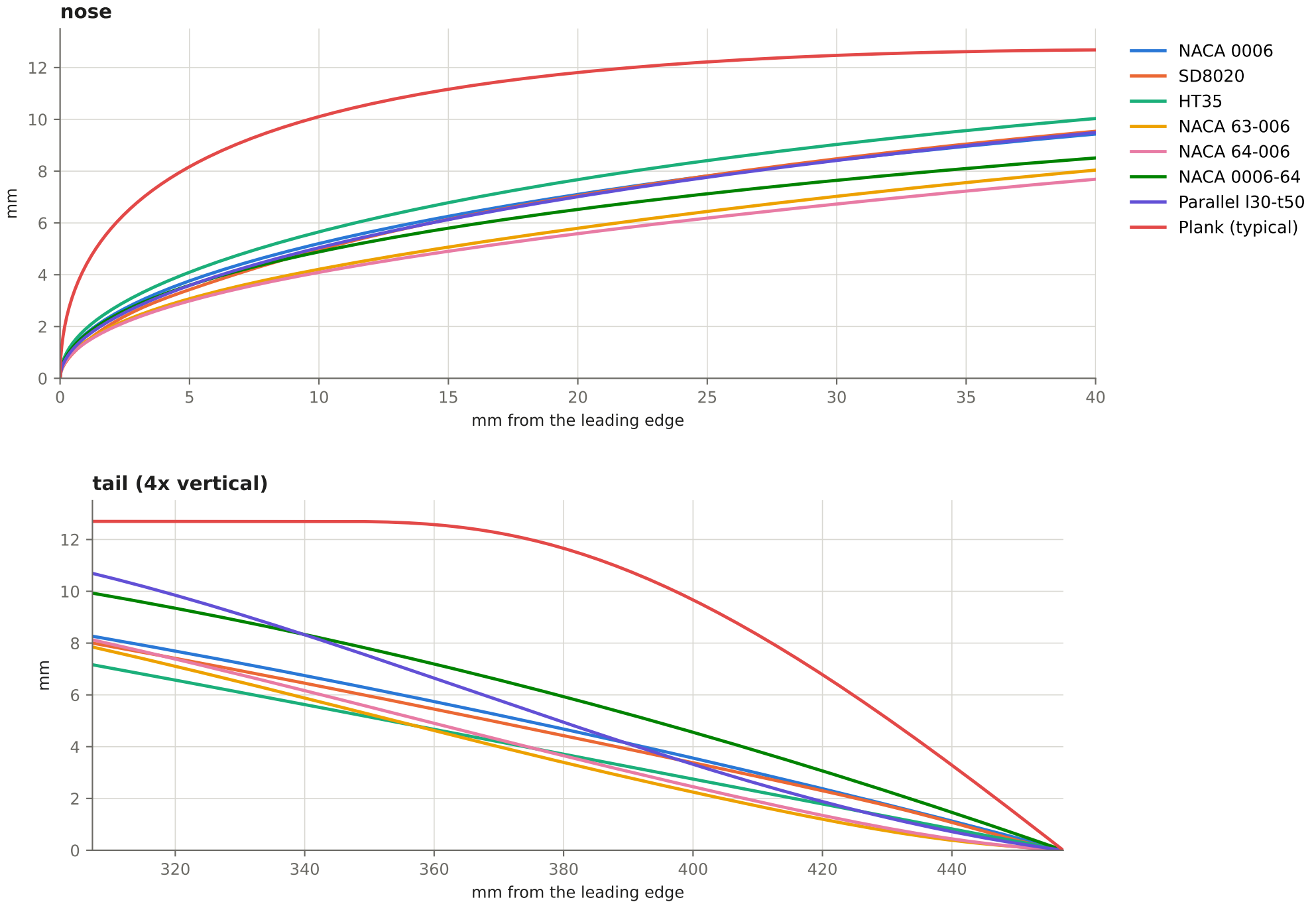
Half-thickness along the chord, against the baseline

Vertical scale stretched about 7x. Grey: NACA 0006 at 1 in. All sections reach 12.7 mm (half of 1 in); where they reach it, and how they get there, differs.



Noses and tails, true scale

Upper surface only (the sections are symmetric). Nose: first 40 mm at true scale. Tail: last 150 mm, vertical scale stretched 4x.



Section properties and the XFOIL estimates

Thickness in mm at stations along the 457 mm chord.

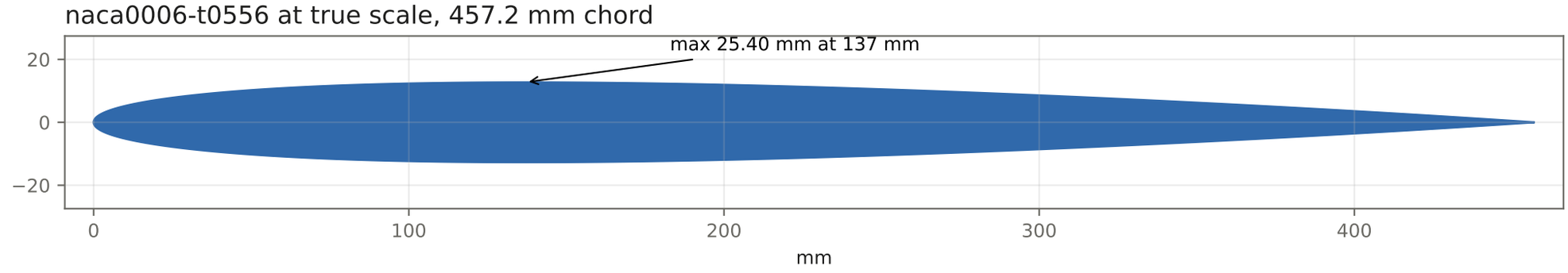
XFOIL estimate of the board's drag against NACA 0006 (negative = less drag), rake 10°: upwind = mean over 1-5 kn at the 1/2/4° loads; downwind = 0° over 1-15 kn.

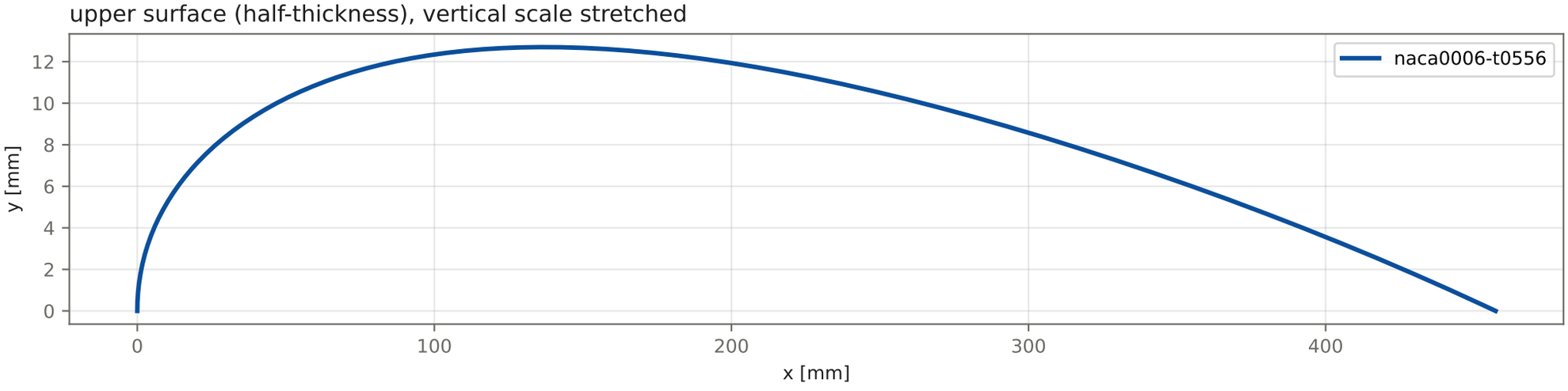
XFOIL under-predicts drag at Re 3e6 by 2-25 % and misses the CFD's leading-edge bubbles; the CFD decides.

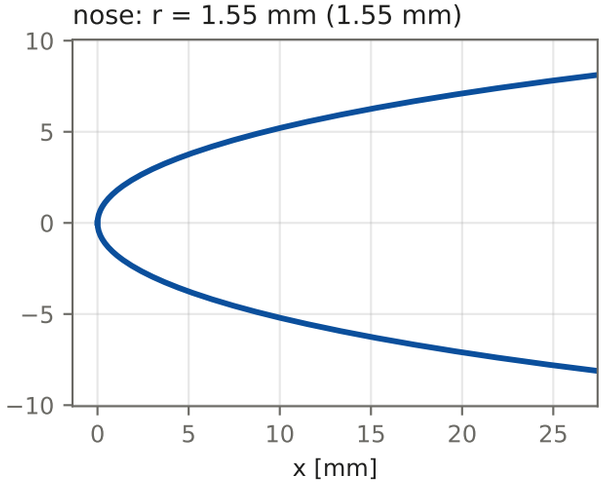
section	max at	nose r	TE	t@5%	t@10%	t@25%	t@50%	t@75%	t@90%	up Tu0.1	up Tu1	dn Tu0.1	dn Tu1
NACA 0006	137 mm	1.5 mm	7.7°	15.0	19.8	25.1	22.4	13.2	5.8	+0.0 %	+0.0 %	+0.0 %	+0.0 %
SD8020	132 mm	0.8 mm	4.9°	15.0	20.1	25.2	22.1	12.6	5.5	-1.0 %	+0.2 %	+1.5 %	+2.2 %
HT35	121 mm	2.0 mm	5.5°	16.2	21.0	25.4	20.9	11.0	4.4	+1.0 %	+0.9 %	+5.7 %	+5.7 %
NACA 63-006	162 mm	1.0 mm	2.2°	12.3	17.1	24.1	23.1	11.4	3.3	+11.9 %	+2.7 %	-1.2 %	-12.8 %
NACA 64-006	172 mm	1.0 mm	1.8°	11.9	16.3	23.5	23.6	12.0	3.6	+10.3 %	+3.6 %	-2.4 %	-15.9 %
NACA 0006-64	183 mm	1.5 mm	10.0°	13.8	17.9	23.7	24.6	16.3	7.4	+4.8 %	+1.7 %	+0.3 %	-5.0 %
Parallel l30-t50	229 mm	1.3 mm	3.9°	14.9	20.0	25.3	25.4	16.2	4.9	+7.0 %	+2.7 %	+7.1 %	+1.1 %
Plank (typical)	343 mm	12.7 mm	22.1°	24.1	25.4	25.4	25.4	25.4	16.2	+41.0 %	+21.5 %	+142.0 %	+57.5 %

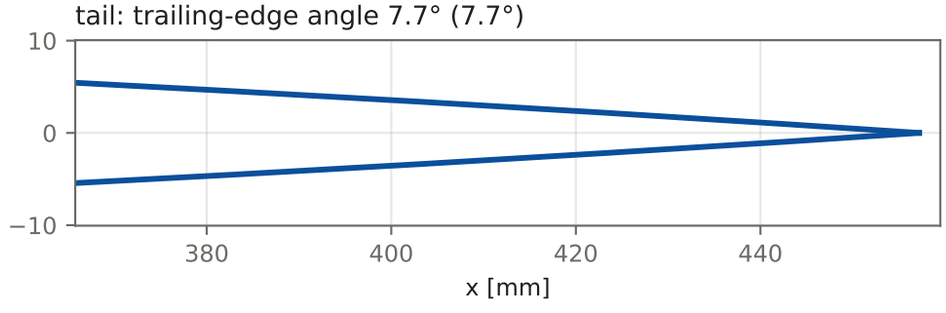
NACA 0006: baseline (user)
SD8020: XFOIL's upwind leader at Tu 0.1 %; sharp nose
HT35: the IOM winner
NACA 63-006: laminar 6-series, best compromise in XFOIL
NACA 64-006: laminar 6-series; validation section
NACA 0006-64: modified 4-digit, crest at 40 %
Parallel l30-t50: best parallel-sided board (XFOIL)
Plank (typical): assumed home-built wood-lead board

NACA 0006: baseline (user)



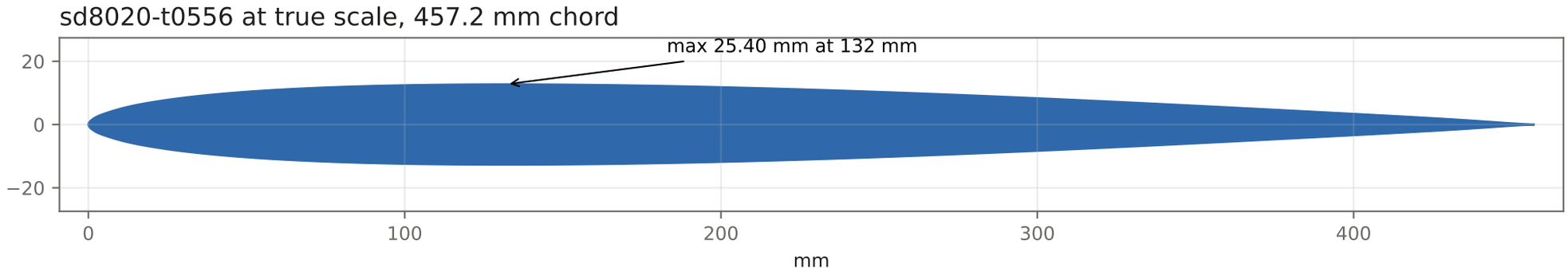


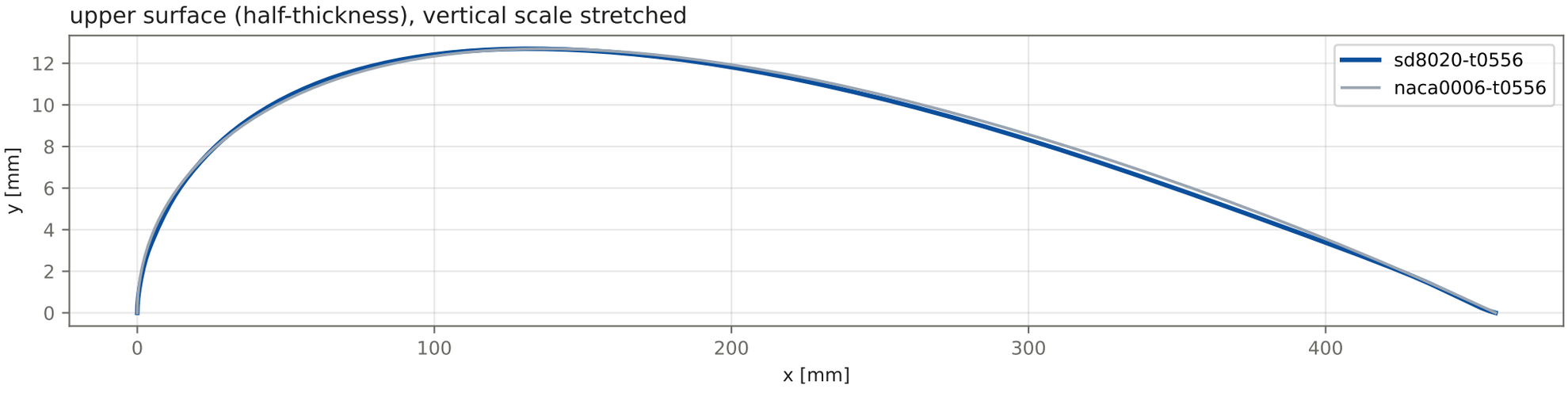


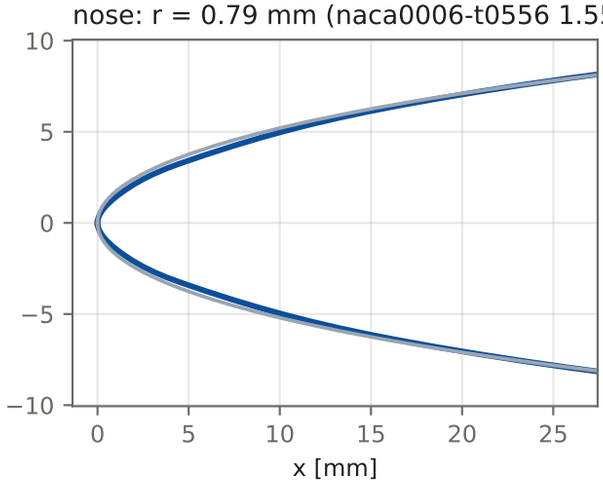


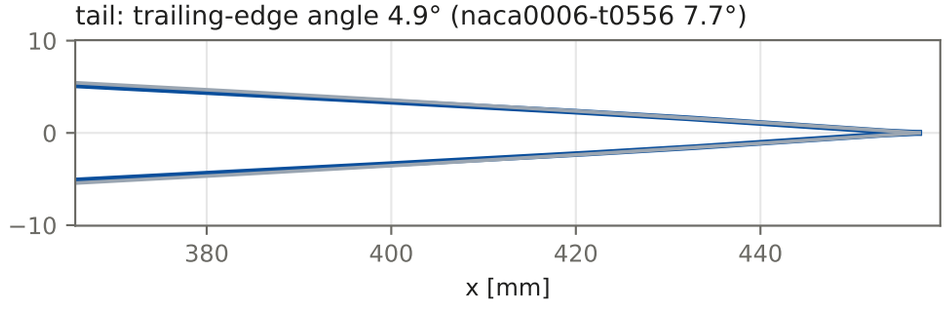
thickness [mm] at x = 23 46 114 229 343 411
naca0006-t0556 15.05 19.82 25.15 22.38 13.21 5.78

SD8020: XFOIL's upwind leader at Tu 0.1 %; sharp nose







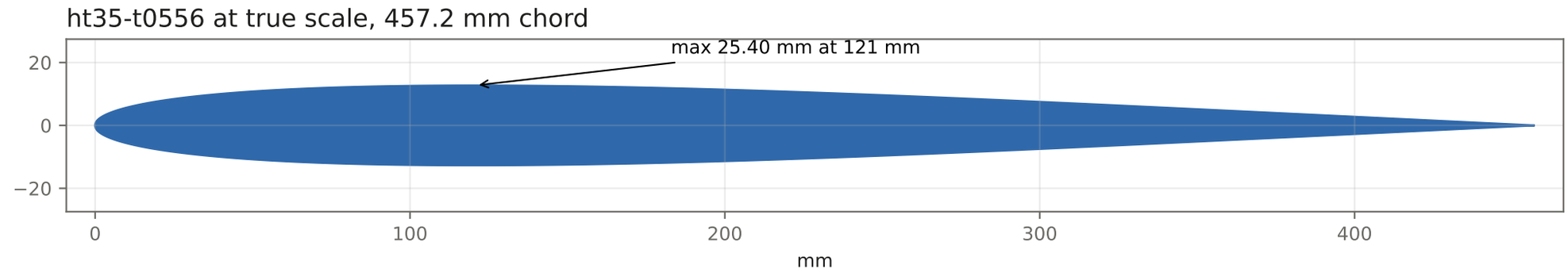


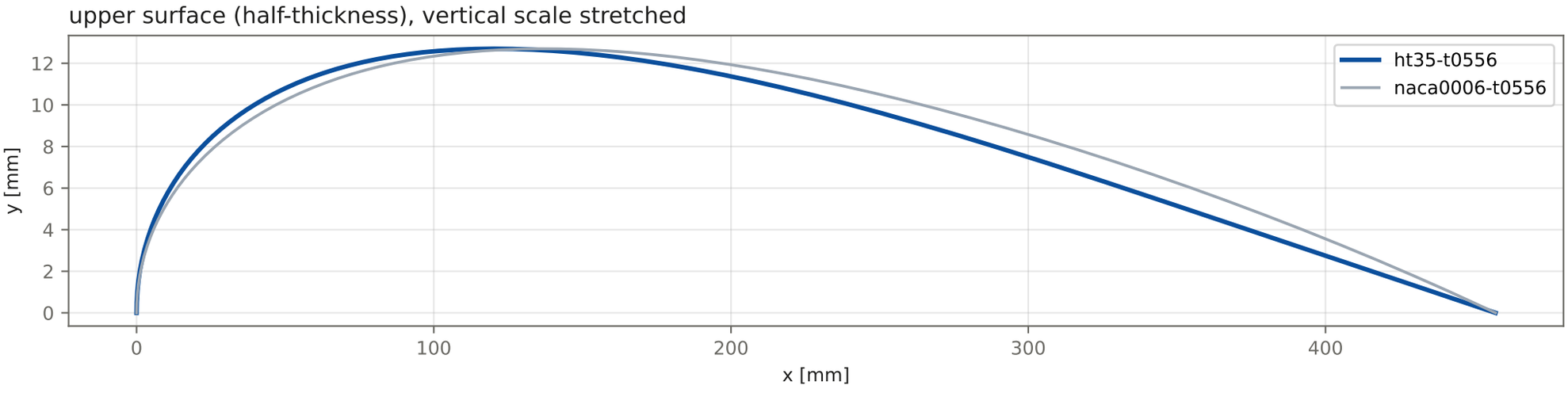
thickness [mm] at x = 23 46 114 229 343 411

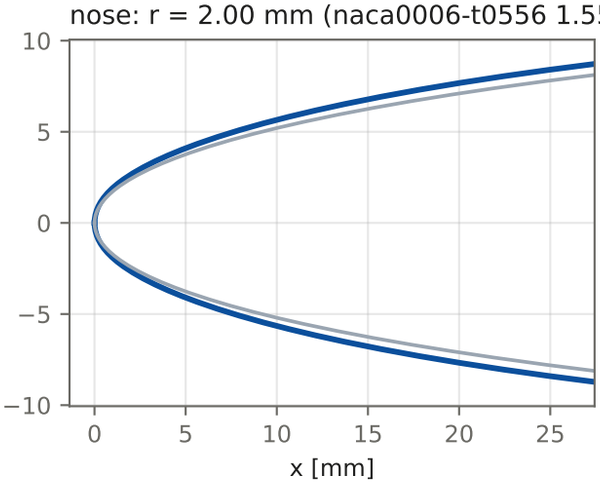
sd8020-t0556	15.01	20.07	25.24	22.06	12.62	5.54
naca0006-t0556	15.05	19.82	25.15	22.38	13.21	5.78

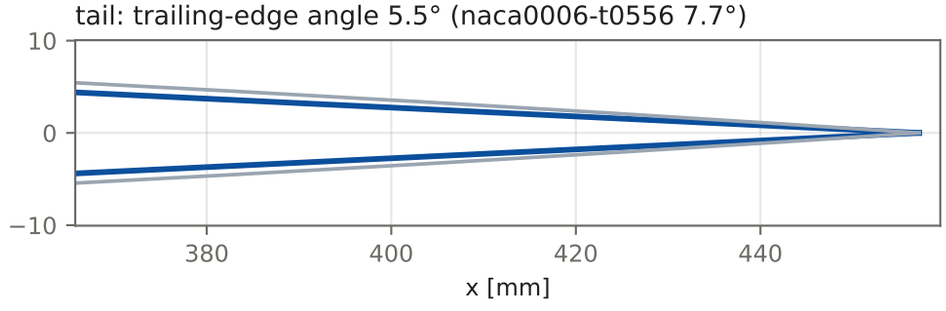
bending stiffness against naca0006-t0556: solid 0.983, skin 0.984

HT35: the IOM winner





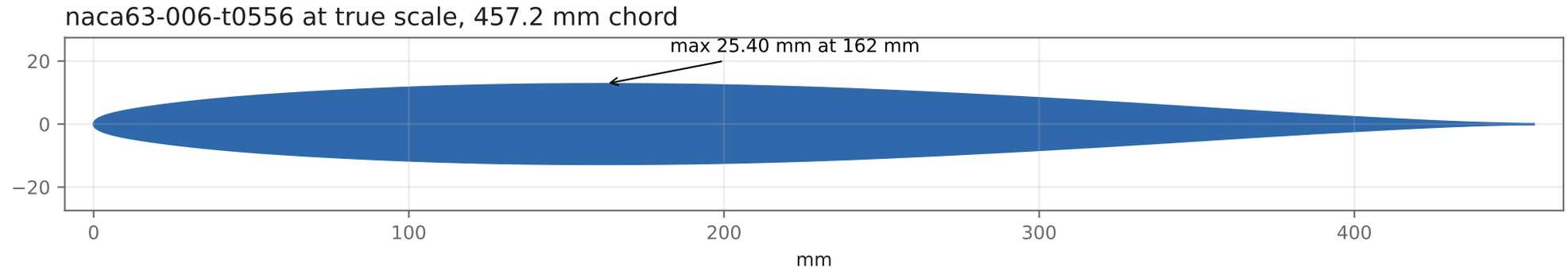


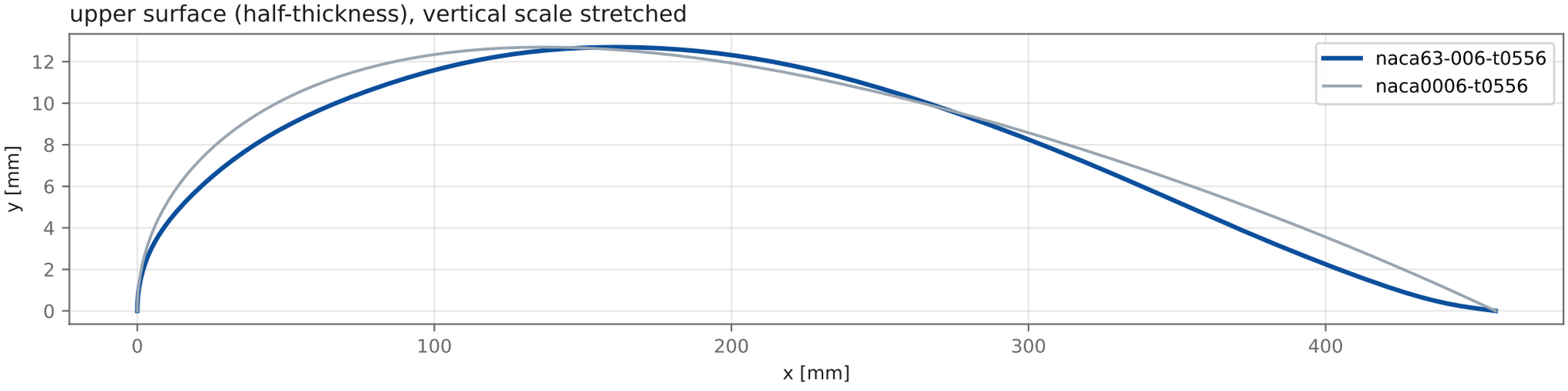


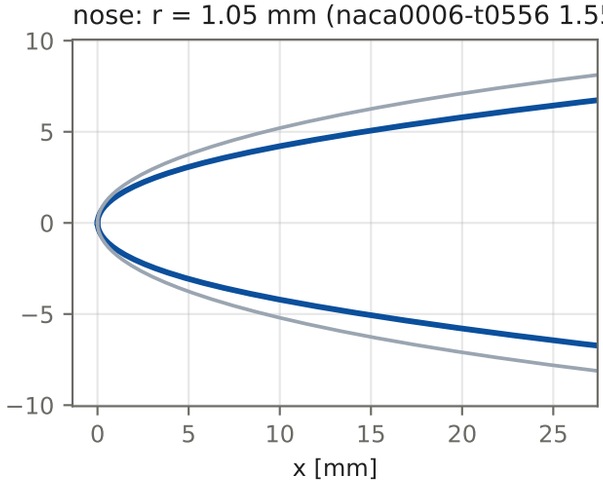
thickness [mm] at x =	23	46	114	229	343	411
ht35-t0556	16.22	20.99	25.38	20.87	10.98	4.40
naca0006-t0556	15.05	19.82	25.15	22.38	13.21	5.78

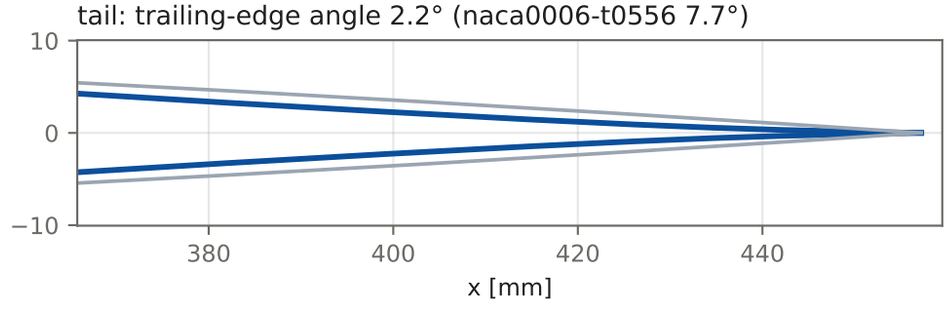
bending stiffness against naca0006-t0556: solid 0.931, skin 0.938

NACA 63-006: laminar 6-series, best compromise in XFOIL





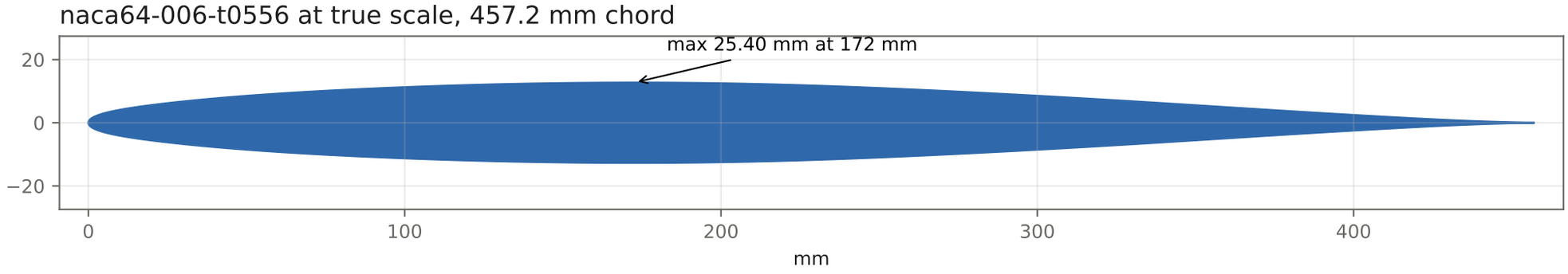


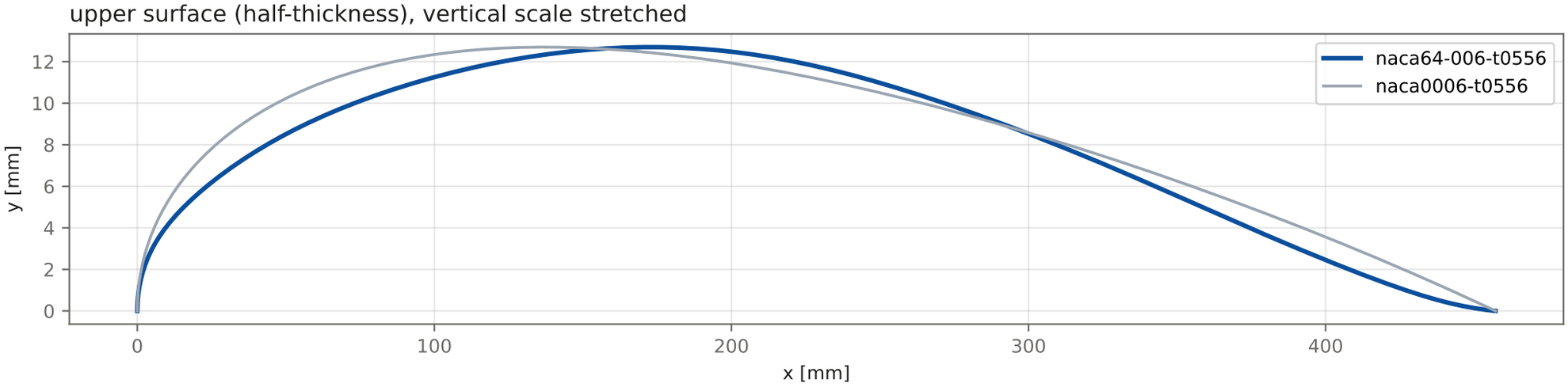


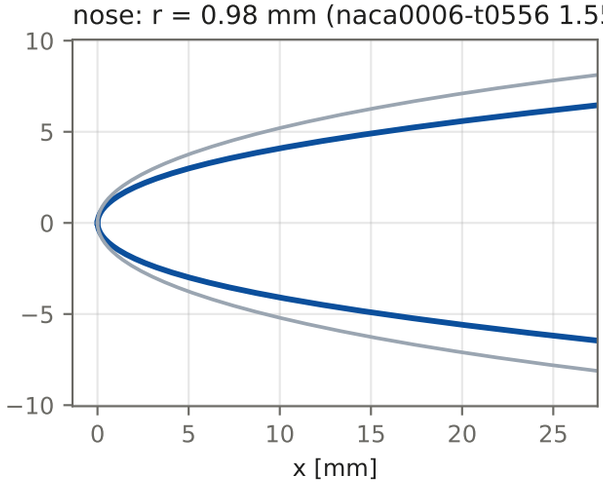
thickness [mm] at x =	23	46	114	229	343	411
naca63-006-t0556	12.35	17.08	24.13	23.10	11.39	3.25
naca0006-t0556	15.05	19.82	25.15	22.38	13.21	5.78

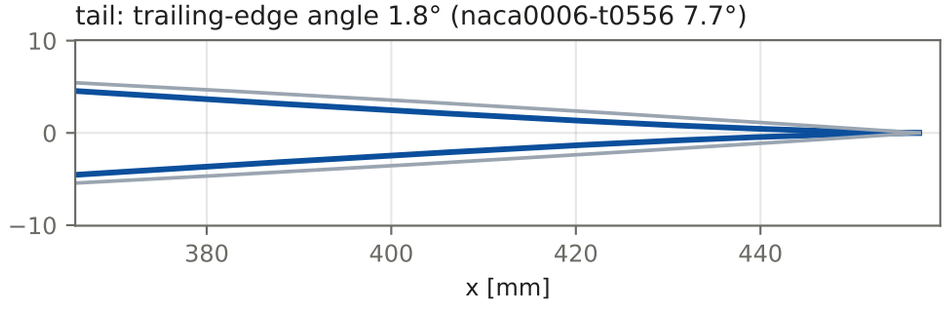
bending stiffness against naca0006-t0556: solid 0.923, skin 0.926

NACA 64-006: laminar 6-series; validation section





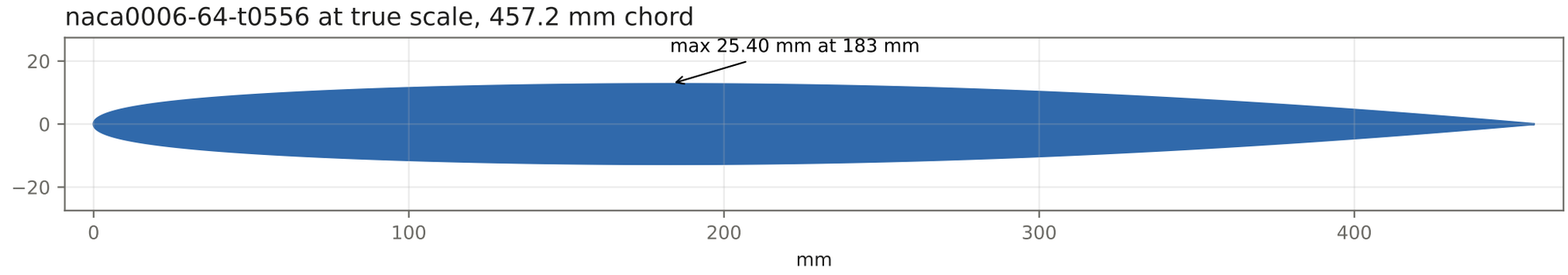


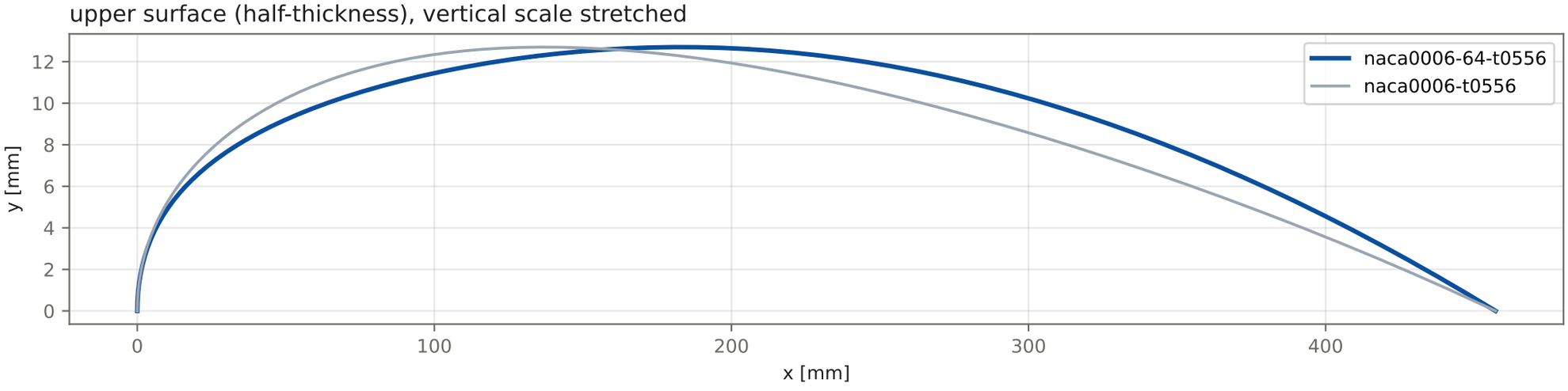


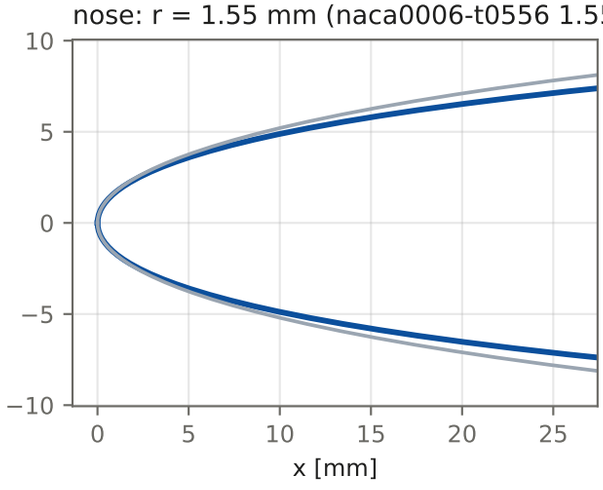
thickness [mm] at x =	23	46	114	229	343	411
naca64-006-t0556	11.88	16.35	23.51	23.55	11.97	3.60
naca0006-t0556	15.05	19.82	25.15	22.38	13.21	5.78

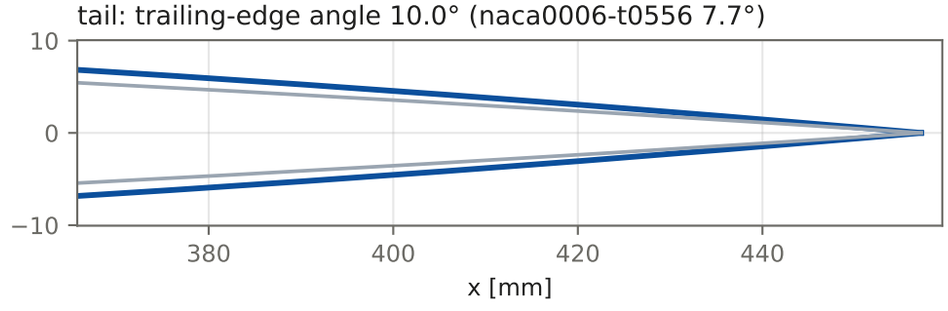
bending stiffness against naca0006-t0556: solid 0.920, skin 0.927

NACA 0006-64: modified 4-digit, crest at 40 %





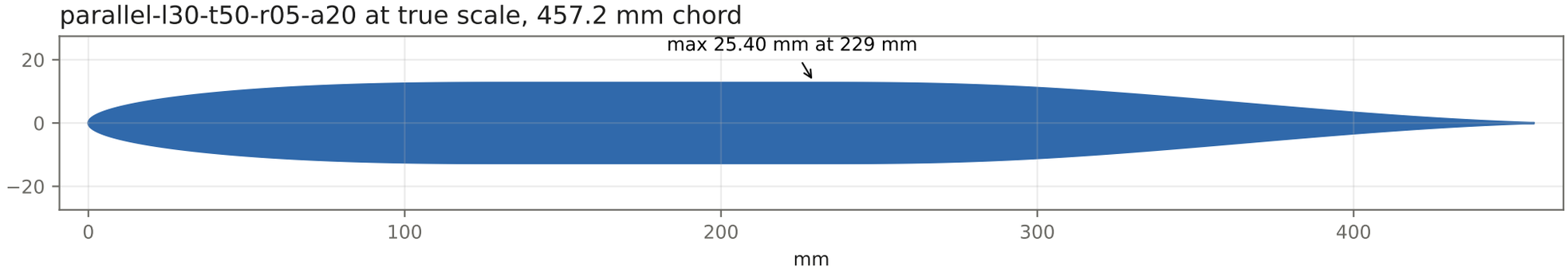


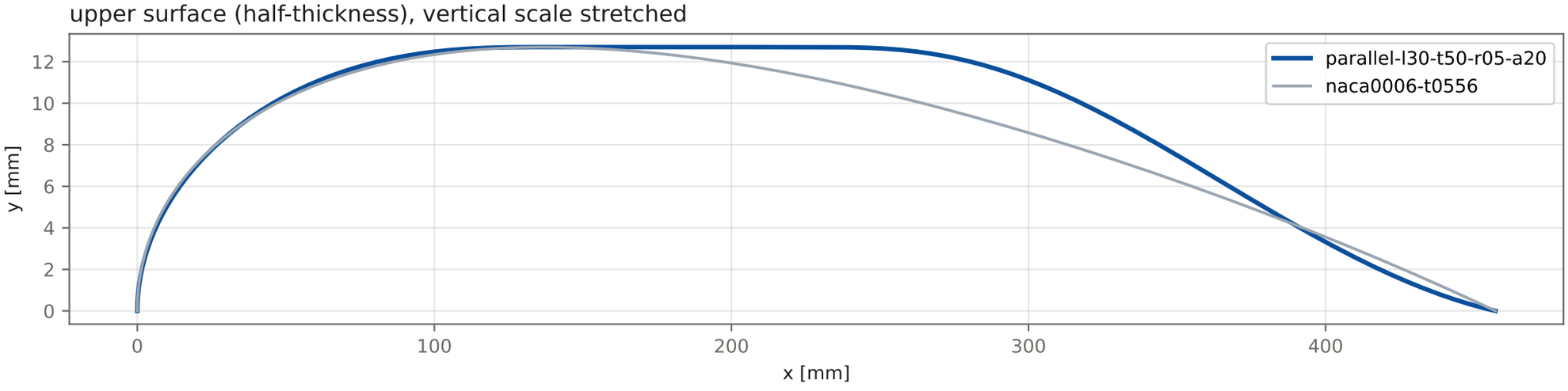


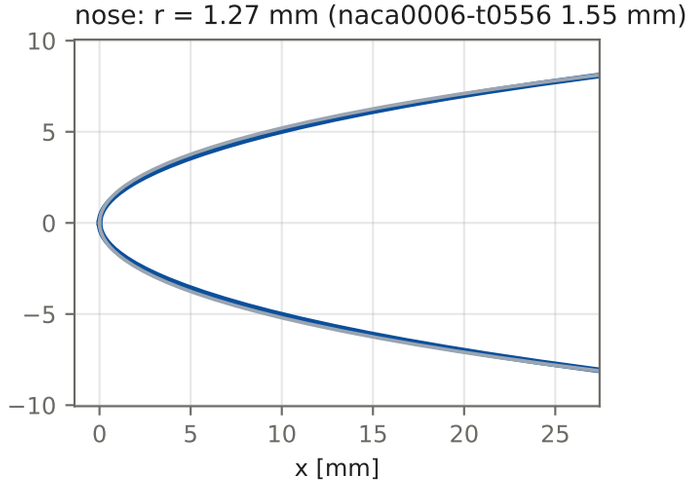
thickness [mm] at x =	23	46	114	229	343	411
naca0006-64-t0556	13.76	17.85	23.69	24.63	16.35	7.43
naca0006-t0556	15.05	19.82	25.15	22.38	13.21	5.78

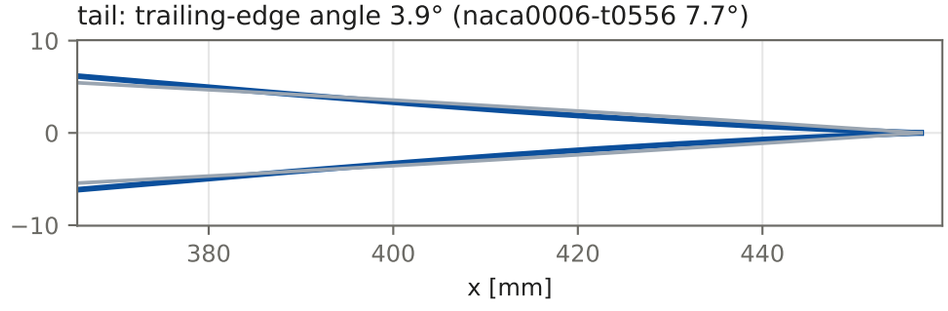
bending stiffness against naca0006-t0556: solid 1.091, skin 1.077

Parallel l30-t50: best parallel-sided board (XFOIL)



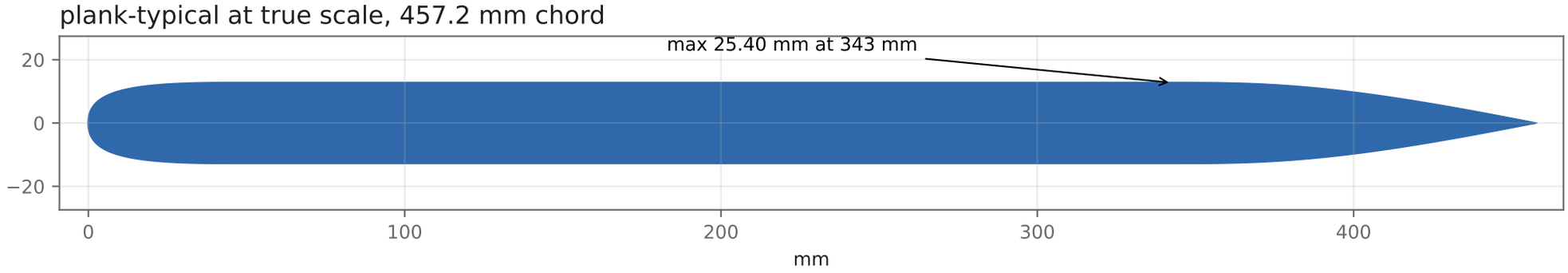


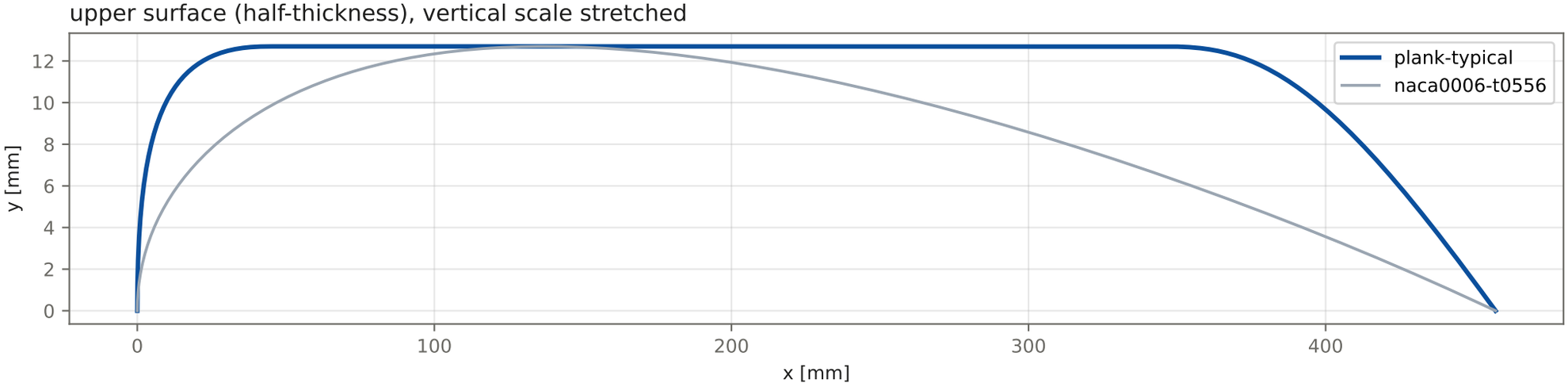


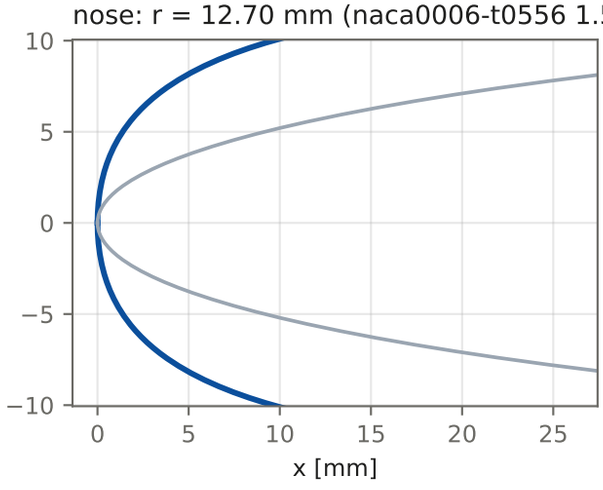


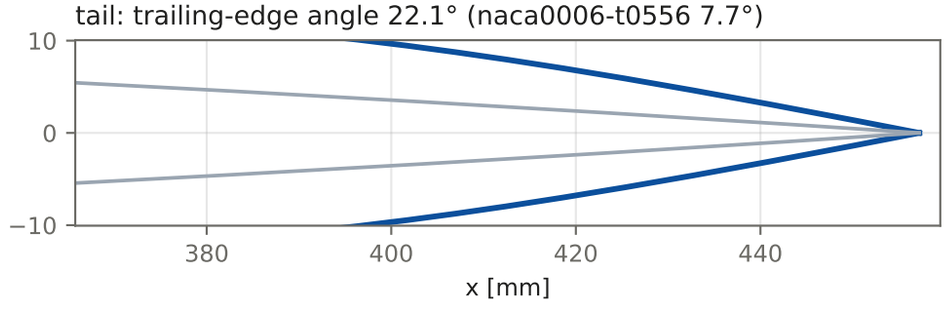

```
thickness [mm] at x =    23    46   114   229   343   411
  parallel-l30-t50-r05-a2014.91  20.00  25.29  25.40  16.17  4.93
  naca0006-t0556      15.05  19.82  25.15  22.38  13.21  5.78
bending stiffness against naca0006-t0556: solid 1.252, skin 1.172
```

Plank (typical): assumed home-built wood-lead board










```
thickness [mm] at x =    23    46   114   229   343   411
  plank-typical       24.13  25.40  25.40  25.40  25.40  16.21
  naca0006-t0556      15.05  19.82  25.15  22.38  13.21  5.78
bending stiffness against naca0006-t0556: solid 1.790, skin 1.582
```